6 Workload-Level Budget Calibration

Unlabelled agreement with a larger-budget reference selects one multiplier for later batches. Reuse must repay calibration at the required quality.

6.1 Why Fixed Refinement Budgets Can Be Wasteful

For routed $U(q)$, the refinement multiplier sets [16]

$$c(\alpha) = \max\{k, \lceil \alpha k \rceil\}, \qquad C_\alpha(q) = \mathrm{Top}^{\mathrm{primary}}_{\min\{c(\alpha),|U(q)|\}} U(q). \tag{6.1}$$

Analysis requires $|C_\alpha| \geq k$; underfull slots are sentinels. Budgets change retention, probes U.

At $k = 10$, the grid retains 20–2,000 survivors without retraining. Lower budgets save reads, not residual storage or previously reserved workspace.

At fixed batch/routing, write

$$T(\alpha) = T_{\mathrm{base}} + T_{\mathrm{select}}(c(\alpha)) + T_{\mathrm{refine}}(c(\alpha)) + T_{\mathrm{final}}(c(\alpha)). \tag{6.2}$$

T_{base} covers transfers/rotation/routing/primary arithmetic (figure 6.1). Selection/resource transitions make latency non-monotone; the rule minimises accepted budget (section 7.5.5).

Trials reuse the index but rerun full searches, including rotation/probing/primary scoring; calibration costs more than residual work alone.

Position among adaptive ANN methods. α serves the candidate-expansion role of Faiss's `k_factor` and cuVS refinement [7, 10]. Alternatives use GT tuning [9], learned per-query stopping [19] or predicted recall [6]. Here unlabelled agreement chooses one reusable workload budget. Scores still use reconstructed vectors, unlike the cuVS source-vector refinement example.

6.2 GT-Free Sampled Output Agreement

Let $\mathcal{R}_\alpha(q)$ be the set of k valid, unique database IDs returned at budget α on a fixed index. Define $\mathcal{R}_{\mathrm{ref}}(q) = \mathcal{R}_{\alpha_{\mathrm{ref}}}(q)$. The descending grid is

$$A = (200, 100, 64, 32, 16, 8, 4, 2), \qquad \alpha_{\mathrm{ref}} = A[0] = 200. \tag{6.3}$$

For S calibration queries $\mathcal{Q}_S$, define

$$\begin{aligned} D_S(\alpha) &= \sum_{q \in \mathcal{Q}_S} \left(k - |\mathcal{R}_\alpha(q) \cap \mathcal{R}_{\mathrm{ref}}(q)|\right) \\ &= \frac{1}{2} \sum_{q \in \mathcal{Q}_S} |\mathcal{R}_\alpha(q) \triangle \mathcal{R}_{\mathrm{ref}}(q)|. \end{aligned} \tag{6.4}$$

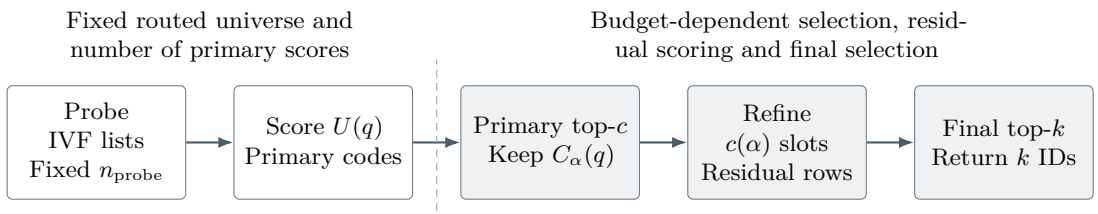


Figure 6.1. The refinement budget changes work after primary-score production. With a fixed query, index and probe count, it leaves the routed candidate universe unchanged. The highlighted stages depend on survivor capacity; this is a logical work decomposition, not a measured latency breakdown. The scan kernel also performs budget-dependent admission and compaction. Padding applies when fewer than c candidates exist.

Accept iff $D_S(\alpha) \leq \epsilon = 1$: at most one missing reference ID, ignoring order/distances. The finite-grid objective is

$$\widehat{\alpha} = \min\{\alpha \in A : D_S(\alpha) \leq \epsilon\}. \tag{6.5}$$

The reference is a fallback. Section 6.3 gives the conditions for exact bisection.

At $k = 10$, $\epsilon = 1$ gives

S	32	64	128	500
$\epsilon/(Sk)$	0.003125	0.0015625	0.00078125	0.0002

Larger samples change coverage and strictness. Disjoint held-out queries evaluate draws from one finite pool (section 7.5).

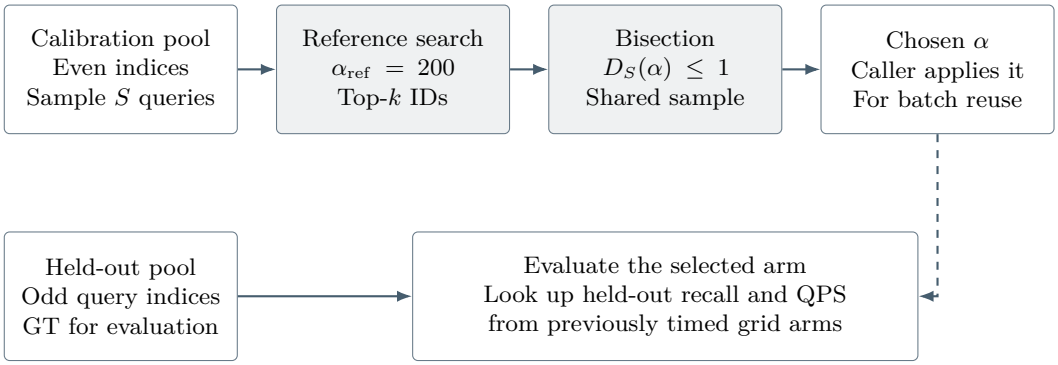


Figure 6.2. Workload-level calibration and its evaluation. Budget decisions use reference IDs; GT assesses held-out quality. Selected-arm timing comes from fixed-arm measurements. Implementation boundaries are in section 3.5.3.

6.2.1 What agreement controls

Under Proposition 6.1, disagreement counts reference answers excluded by primary screening:

$$D_S(\alpha) = \sum_{q \in \mathcal{Q}_S} |\mathcal{R}_{\text{ref}}(q) \setminus C_\alpha(q)|. \tag{6.6}$$

For k-element outputs and GT set $\mathcal{T}_k$, set difference gives

$$|\mathcal{R}_{\text{ref}} \cap \mathcal{T}_k| - |\mathcal{R}_\alpha \cap \mathcal{T}_k| \leq |\mathcal{R}_{\text{ref}} \setminus \mathcal{R}_\alpha|.$$

Summing and dividing by Sk yields the relative bound. On the same sample,

$$\begin{aligned} r_{\mathcal{Q}_S}(\alpha) &\geq r_{\mathcal{Q}_S}(\text{ref}) - \frac{D_S(\alpha)}{Sk}, \\ D_S(\alpha) \leq \epsilon \implies r_{\mathcal{Q}_S}(\alpha) &\geq r_{\mathcal{Q}_S}(\text{ref}) - \frac{\epsilon}{Sk}. \end{aligned} \tag{6.7}$$

Scope. The bound is relative to the reference on this sample. GT evaluation measures the reference's remaining error (section 7.8.2).

6.2.2 Sampling resolution and ordered testing

Let H_S count queries with any missing reference ID. Since $H_S \leq D_S$, acceptance implies $H_S \leq 1$. For a prespecified budget and i.i.d. queries, the one-event one-sided 95% binomial limit solves [22]

$$(1-u)^S + Su(1-u)^{S-1} = 0.05. \tag{6.8}$$

The model-dependent resolutions are

	$S = 128$	$S = 500$
i.i.d. query events	0.03652	0.009452
i.i.d. homogeneous result slots	0.003701	0.0009484

The stronger slot model gives approximately $4.74/(Sk)$. Query i.i.d. does not imply slot independence. Neither model certifies fixed-pool draws.

Monotonicity permits fixed-sequence control only with valid per-arm tests [3]. Our one-slot threshold measures agreement, not certified GT risk.

A finite-pool extension. Assume fixed per-query $X_q(\alpha)$, independent of subset/position within the execution regime. For pool size N_{cal} and counts $n_j = |\{q : X_q = j\}|$, uniform sampling without replacement accepts only all-zero samples or one single-count query [23]:

$$\Pr\{D_S(\alpha) \le 1\} = \frac{\binom{n_0}{S} + n_1\binom{n_0}{S-1}}{\binom{N_{\text{cal}}}{S}}. \tag{6.9}$$

Out-of-range binomial coefficients are zero. Predicting GT risk also needs the grid acceptance structure and held-out fixed-arm quality. Recorded subset sums check this empirically (section 7.5.5).

6.3 Budget Search, Nestedness and Tie Safety

6.3.1 Bisection and complete enumeration

Write $J = |A|$. The implementation computes the sample reference once, then searches for the largest acceptable index in the descending grid:

1. Set $l = 0$, $h = J - 1$. Zero is the fallback.
2. While $l < h$, test $j = \lfloor (l + h + 1)/2 \rfloor$ by running $A[j]$ and computing $D_S(A[j])$.
3. If $D_S(A[j]) \le \epsilon$, set $l = j$; otherwise set $h = j - 1$.
4. Return $A[l]$. Apply it before reuse.

The eight-point grid needs at most three complete candidate searches in addition to the reference. Section 3.5.3 specifies invocation/timing.

Enumeration reruns reference plus seven arms on the same sample. It handles acceptance holes; bisection finds its minimum only for prefix acceptance. Both timers include the reference.

6.3.2 A sufficient condition for monotonicity

Proposition 6.1 (Nested survivors preserve reference membership)**.**
For one query, assume fixed routing and survivor sets $C_\alpha \subseteq C_\beta \subseteq C_{\alpha_{\text{ref}}}$ whenever $\alpha \le \beta \le \alpha_{\text{ref}}$. Assume at least k valid survivors, fixed refined scores and one deterministic total order for final ties. Each result is the exact top-k in that order. Then

$$|\mathcal{R}_\alpha \cap \mathcal{R}_{\text{ref}}| = |C_\alpha \cap \mathcal{R}_{\text{ref}}|, \qquad D_S(\alpha) \ge D_S(\beta). \tag{6.10}$$

The latter statement holds when these conditions hold for every sampled query.

Proof.
For $x \in \mathcal{R}_{\text{ref}} \cap C_\alpha$, the common total order gives

$$|\{y \in C_\alpha : y \prec x\}| \le |\{y \in C_{\alpha_{\text{ref}}} : y \prec x\}| < k.$$

Hence $x \in \mathcal{R}_\alpha$. Together with $\mathcal{R}_\alpha \subseteq C_\alpha$, this proves the intersection identity in equation (6.10). Nestedness gives $|C_\alpha \cap \mathcal{R}_{\text{ref}}| \le |C_\beta \cap \mathcal{R}_{\text{ref}}|$. Summing missing reference counts over queries proves $D_S(\alpha) \ge D_S(\beta)$. □

Fixed-key exact primary top-c gives nesting; fixed-key refinement then gives monotone agreement. This supports bisection, without making GT recall or runtime monotone as the pool grows.

6.3.3 A corollary for the evaluated grouped path

Within a draw, S, routing and scan threads are fixed; `scheduled_groups(S)` ignores c. Grouped budgets use the same LUT sums. Fixed B_r, 16 lanes and final $(D_{\mathrm{H}}, \mathrm{ID})$ order keep candidate scores fixed.

Corollary 6.2 (Grouped calibration)**.**
For the evaluated grouped configuration, under the representation and shared-bound contracts of Chapters 4 and 5, primary survivors are nested across budgets. Consequently Proposition 6.1 applies and bisection returns the same minimum accepted grid budget as enumeration.

Proof.
Lemma 5.1 and propositions 5.1 and 5.2 give exact primary top-c under the common key $(D_{\mathrm{P}}, \mathrm{position})$. Its prefixes nest as c grows. Fixed refinement/final order supplies proposition 6.1's remaining premises. □

The corollary covers 2,176 draws; 128 OpenAI-3072/S = 128 draws use distance-only selection. All 2,304 recorded curves are monotone and match enumeration (section 7.5.5); this is finite evidence.

Full-pool coverage. Two FULL cells group Vogue; ten use $g_{\mathrm{scan}} = 0$. All twelve match enumeration. Ten distance-only cells remain outside corollary 6.2: nesting needs a common cutoff order or no crossing ties. A matching held-out path does not ensure this.

A rank-based extension. Under the same nesting/fixed-score premises, one-based primary ranks give

$$D_S(\alpha) = \sum_{q \in \mathcal{Q}_S} \sum_{x \in \mathcal{R}_{\mathrm{ref}}(q)} \mathbf{1}\{\mathrm{rank}_{\mathrm{P}}(x) > c(\alpha)\}. \tag{6.11}$$

Exporting reference ranks could replace repeated budget searches. This is unimplemented.

6.4 Budget Baselines and Their Questions

Table 6.1 separates budget selection from its evaluation. Fixed arms share the index, route and held-out queries; invocation and timing details are in section 3.5.3.

For held-out recall/QPS $\widehat{r}, \widehat{Q}$ and $\widehat{r}^* = \max_{\alpha \in A} \widehat{r}(\alpha)$,

$$\alpha_{\mathrm{oracle}} \in \arg\max_{\alpha \in A} \left\{ \widehat{Q}(\alpha) : \widehat{r}(\alpha) \geq \widehat{r}^* - \delta \right\}, \qquad \delta = 10^{-3}. \tag{6.12}$$

This labelled/timed diagnostic uses recall slack δ, not slot allowance ϵ.

FULL tightens $\epsilon/(Sk)$ without labels; enumeration checks search correctness. Agreement minimises accepted budget, while the oracle maximises quality-constrained speed.

Fixed 64/200 test smaller/high budgets (section 7.5 and table 7.9).

Case	C_1	C_2	C_3	Reference overlap
Nested	$\{b\}$	$\{b, c\}$	$\{a, b, c\}$	0, 0, 1
Non-nested	$\{a\}$	$\{b, c\}$	$\{a, b, c\}$	1, 0, 1

Figure 6.3. Constructed $k = 1$ example with refined order $a \prec b \prec c$ and reference $\{a\}$. The nested case preserves reference membership. With all primary scores tied, different tie choices can produce the non-nested middle pool and break monotonicity. These are explanatory sets, not measured GPU outputs.

Table 6.1. Budget controls and the questions they answer. Held-out GT is used to assess every result, but only the diagnostic oracle uses it to choose a budget. The reference is fixed at 200 for agreement-based arms.

Control	Selection information	Question
Fixed 100	No calibration	Gain relative to the inherited default?
Fixed 64 and smaller grid arms	No calibration	Does a simple fixed choice remove the need for calibration?
Sampled bisection	S unlabelled queries; slot agreement	How well does a low-cost sample choose a reusable budget?
Same-sample enumeration	Same S, reference and threshold; all grid arms	Did bisection miss the smallest accepted grid budget?
Full-pool calibration	All 500 calibration queries; same bisection rule	What changes with broader coverage and a stricter normalised allowance?
Held-out oracle	GT recall and timed QPS for all arms	Which measured arm is fastest under the diagnostic quality constraint?

6.5 Hold-out Risk, Calibration Cost and Payback

6.5.1 Choosing a calibration scale

The strong independent-slot model suggests

$$\frac{4.74}{Sk} \simeq 10^{-3}, \quad k = 10 \quad \Longrightarrow \quad S \simeq 474 \approx 500.$$

This resolves reference disagreement, not GT risk (section 6.2). FULL quality and sampled risks are in table 7.8 and figure 7.6.

FULL still needs the path/order conditions in section 6.3.

6.5.2 Risk and selected-arm performance

Over m calibration draws, let draw r select $\widehat{\alpha}_r$. Define held-out loss and threshold-crossing frequency by

$$L_r = \widehat{r}^* - \widehat{r}(\widehat{\alpha}_r), \qquad \widehat{p}_\delta = \frac{1}{m}\sum_{r=1}^{m} \mathbf{1}\{L_r > \delta\}. \tag{6.13}$$

Mean loss, crossing frequency and tail size answer different questions. Table 7.9 and figure 7.6 report the archived summaries; the largest supplementary per-draw losses are reported in section 7.5.5.

Reported selected-arm rate is $\overline{Q} = m^{-1}\sum_r \widehat{Q}(\widehat{\alpha}_r)$, a lookup from timed fixed arms. It is distinct from the median budget's rate and from the throughput of a calibrated query stream. Quality and speed must be read together (table 7.10).

6.5.3 Cost of reuse

For B_{reuse} equal-sized subsequent batches, model

$$T_{\text{policy}}(B_{\text{reuse}}) = T_{\text{cal}} + B_{\text{reuse}}T_{\text{chosen}}, \qquad T_{\text{fixed}}(B_{\text{reuse}}) = B_{\text{reuse}}T_f. \tag{6.14}$$

When $T_{\text{chosen}} < T_f$, the break-even count is

$$B^*_{\text{reuse}} = \frac{T_{\text{cal}}}{T_f - T_{\text{chosen}}}. \tag{6.15}$$

Indeed,

$$\begin{aligned} T_{\text{policy}} < T_{\text{fixed}} &\iff T_{\text{cal}} < B_{\text{reuse}}(T_f - T_{\text{chosen}}) \\ &\iff B_{\text{reuse}} > B^*_{\text{reuse}} \qquad (T_{\text{chosen}} < T_f). \end{aligned}$$

If $T_{\text{chosen}} \geq T_f$, there is no throughput payback. Batch time is $T = 1000n/Q$ milliseconds. For varying choices, convexity gives $1/\mathbb{E}[Q] \leq \mathbb{E}[1/Q]$: inverse mean QPS understates mean time. Boundaries/direct checks follow sections 3.5.3 and 7.5.6. Campaign estimates are in table 7.10.

Periodic calibration adds T_{cal}/H per batch. Automatic drift detection/renewal is future work.

6.6 Policy Failure Modes and Applicability

Reuse needs stable index/k/routing/queries (section 6.3); quality/tails/cost follow section 6.5. Drift detection is unimplemented.

The caller must explicitly install α before reuse.

Budget evidence uses v69/OpenAI TF32; FP32 frontier checks do not recalibrate it.

7 Experimental Evaluation

We test construction/capacity (RQ1), execution (RQ2), budget quality/reuse (RQ3) and competitive regions (RQ4).

7.1 Experimental Protocol

7.1.1 Comparison protocol and reproducibility

Sections 3.2, 3.3 and 3.5 define the protocol. Search excludes construction/calibration; each ablation has its own baseline.

CPU JHQ uses unmatched settings/protocols: October Vogue has 1,024 lists and $[8, 4]$ bits versus E1's 4,096 lists. A matched CPU/GPU speedup is unavailable.

Adaptive calls/device-only timing are in section 3.5.2.

Metrics and frontier estimators follow sections 3.2 and 3.4; g_{LUT} and g_{scan} retain the distinct meanings in section 5.1.

7.2 Modern GPU ANN Recall–QPS Frontiers

7.2.1 Comparison scope and frontier construction

Figure 7.1 compares JHQ-GPU with the graph, compressed-IVF and brute-force baselines of section 3.4 [17, 28, 33], using the first four datasets in table 3.2. E1 uses batch 1,000/common GT; T5 covers small batches (section 7.2.6), with internal ablations in section 7.6.

E1 dispatches 105 arXiv/OpenAI cells to $g_{\mathrm{scan}} = 0$ and 27 Vogue cells to $g_{\mathrm{scan}} = 4$, triggered by CV 0.338–0.622. Only Vogue exercises grouped retention/bounds/merging (section 5.5).

E1 has 1,590 valid points: JHQ 27–39/dataset, baseline families 54–264. No failure/invalid-ID/duplicate-row flags occur. Sweeps share indexes (section 3.4).

Table 3.3 and section 3.4 define finite coverage; targets beyond it remain unmeasured.

Storage differs: nine-bit RaBitQ shares $9d/8$ nominal code bytes; 4/8-bit codes use less. Table 4.1 and section 7.7 give representation costs and measured allocations alongside this performance envelope.

7.2.2 Matched recall and opposing operating regions

Table 7.1 reports recall 0.97 using the interpolated and qualifying-measured estimators defined in section 3.4.

Removing grouping lowers Vogue's JHQ/RaBitQ ratio at recall 0.97 from 1.059 to 0.933 at fixed index/probes/budget/final outputs (batch 1,000). This isolates the integrated grouping path, not bounds alone.

CAGRA int8 leads 1.20–1.98× at 0.90–0.92, without 0.97 coverage. RaBitQ wins some 0.98+ targets; CAGRA fp32 wins near 0.99 on arXiv (figure 7.1).

7.2.3 Precision sensitivity on the OpenAI datasets

Supplementary source inspection establishes that the E1 implementation uses TF32 multiplication with FP32 accumulation for OpenAI rotation and centroid GEMMs, despite FP32

Table 7.1. Recall 0.97 comparison. Each ratio is interpolated / measured; values above one favour JHQ. JHQ kQPS is the interpolated estimate. The small arXiv–RaBitQ difference changes direction with the estimator.

Dataset	JHQ kQPS	vs CAGRA fp32	vs IVF-RaBitQ
Vogue	230.8	1.416 / 1.537	1.059 / 1.046
arXiv	258.6	1.317 / 1.422	0.996 / 1.015
OpenAI-1536	126.7	1.747 / 2.294	1.130 / 1.111
OpenAI-3072	55.9	1.747 / 2.088	1.157 / 1.346

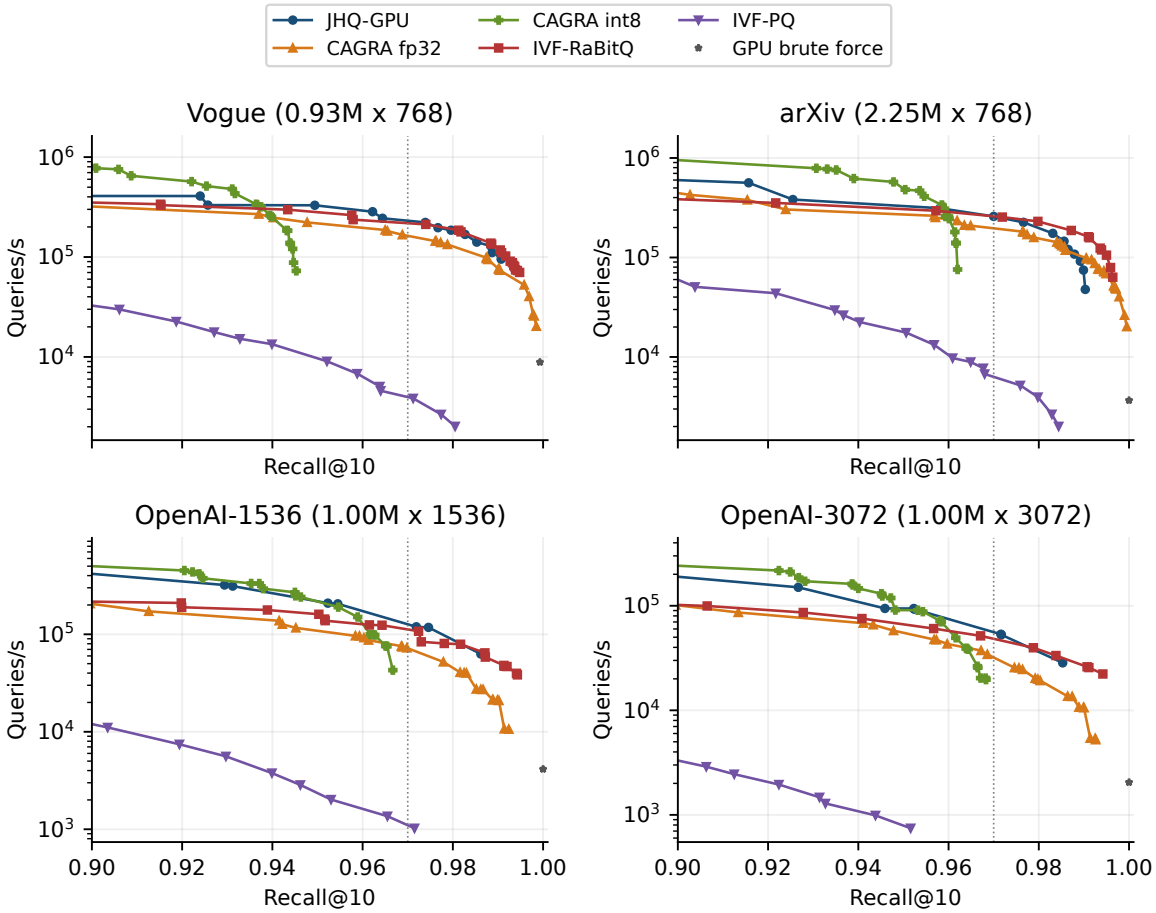


Figure 7.1. Measured host-to-host recall–QPS frontiers on the four E1 datasets. The displayed recall window begins at 0.90, with a vertical line at 0.97; all baseline families, including CAGRA int8, participate in frontier construction. Lines connect measured points; throughput uses a logarithmic axis.

input/storage and a misleading source comment (section 4.2). The main E1 frontier remains the historical measurement of that implementation. In campaign D, a separate FP32 variant changes the cuBLAS math-mode switch and replays 37/39 OpenAI configurations at batch 1,000, five rounds each. Execution order alternates between variants across probe settings. Both reuse the same TF32-trained model; the intervention changes add/query arithmetic, including database encoding and routing. It is not a comparison of independently FP32-trained models. Table 7.2 summarises the paired changes.

Table 7.2. FP32 sensitivity. QPS ratio is FP32/TF32 at equal configuration (geometric mean); recall differences have the same direction. The last two columns compare the FP32 frontier at recall 0.97 with archived E1 IVF-RaBitQ and CAGRA fp32 frontiers. Campaign D does not remeasure the baselines.

Dataset	Pairs	QPS ratio	ΔRecall	vs RaBitQ	vs CAGRA
OpenAI-1536	37	0.953	+0.0003–+0.0005	1.086	1.679
OpenAI-3072	39	0.951	+0.0000–+0.0003	1.079	1.629

At equal configuration, FP32 reduces throughput by about 5% and increases recall by 0–0.0005. At recall 0.97, FP32 is about 3% slower than same-session TF32, yet 1.086/1.079× archived RaBitQ and 1.679/1.629× archived CAGRA fp32. New-session v69 gives 1.118/1.114 against archived RaBitQ, versus historical 1.130/1.157. D compares baselines across sessions; only v69/v72 share a session. Figure 7.2 shows the two JHQ frontiers with their archived

comparators. T1 separately remeasures same-session anchors, with grid-sensitive results in section 7.2.4.

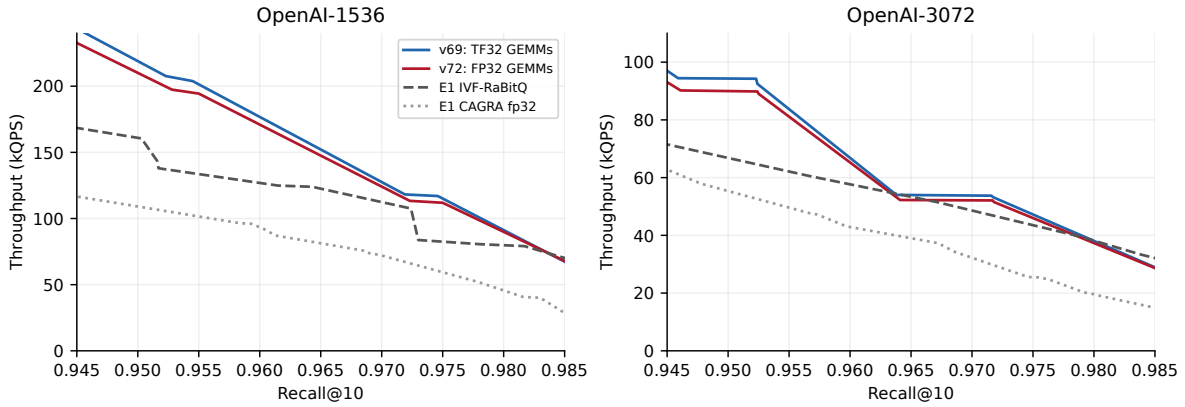


Figure 7.2. Campaign D precision frontiers. Solid curves are same-session JHQ variants; dashed/dotted baselines are historical E1 measurements. All use the chapter's frontier and interpolation rules.

7.2.4 Same-session baseline anchors

T1 uses E1 pools and twelve paired configurations per OpenAI dataset. Two five-round JHQ repetitions bracket baseline runs; medians combine geometrically. Equal-configuration FP32/TF32 means are 0.956/0.964. Both reuse TF32-trained caches, changing add/query arithmetic.

Table 7.3. T1 at recall 0.97, batch 1,000. Baseline ratios are interpolation / strict measured threshold; FP32/TF32 is interpolated. All methods share a session, unlike D's archived baselines.

Dataset	FP32/TF32 at 0.97	FP32 / RaBitQ	FP32 / CAGRA
OpenAI-1536	0.980	1.126 / 1.053	1.752 / 1.633
OpenAI-3072	0.973	1.328 / 1.316	1.825 / 2.019

The OpenAI-3072 RaBitQ bracket (0.9669–0.9796 recall) shows grid sensitivity. These anchors neither replace E1 nor provide FP32 retraining.

7.2.5 Interpretation and comparison limits

Both OpenAI datasets use Qdrant DBpedia embeddings [31, 32]. After truncation/renormalisation, 1,000 query and 10,299 sampled base rows have cosine ≥ 0.9980 (median 1.000000), with GT overlap 9.046/10. Without entity IDs, these checks support row alignment. Doubling width leaves CAGRA ratios at 1.747 and RaBitQ at 1.130/1.157, so width alone is insufficient.

Gains accompany smaller 128-probe anchors: OpenAI 4–8, Vogue 64, arXiv 100 (later FULL 200). This is an association, not an isolated effect or matched-frontier setting.

Section 7.8 checks sensitivity to small margins and the interface/memory/compilation differences of section 3.3.

7.2.6 Small batches change the competitive region

T5 compares scheduled JHQ, CAGRA fp32/int8 and IVF-RaBitQ at batches 8/32 on the four E1 datasets. Five synchronous host-to-host rounds use 1,000/992 queries, respectively. The 1,808 points include a JHQ one-block control, excluding IVF-PQ/brute force. Figure 7.3 and table 7.4 compare the resulting regions.

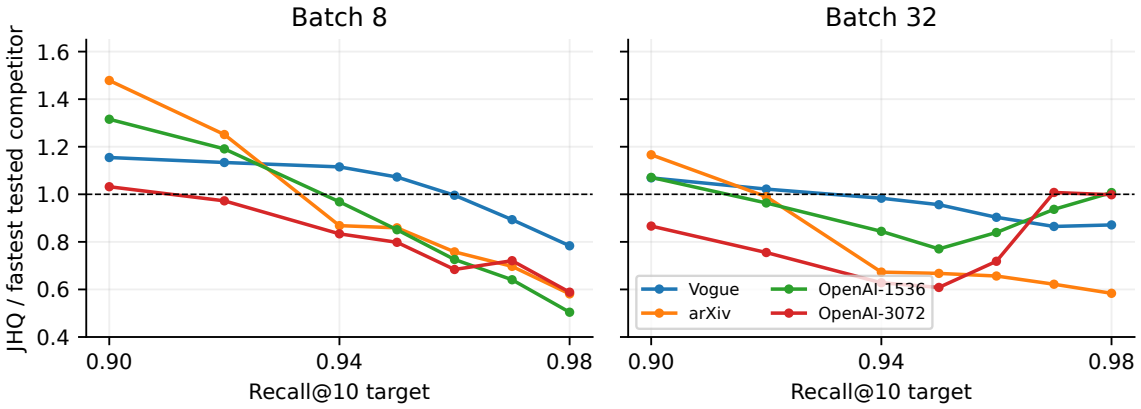


Figure 7.3. T5 JHQ / fastest-competitor QPS under interpolation with qualifying measured endpoints. Above one favours JHQ. Table 7.4 gives strict thresholds.

Table 7.4. T5 QPS ratios: interpolation with qualifying endpoints / strict measured threshold. The fastest competitor is selected independently for each estimator; higher-recall qualifying endpoints remain eligible without extrapolation.

Dataset	b	$R = 0.90$	$R = 0.95$	$R = 0.97$
Vogue	8	1.155 / 1.091	1.073 / 0.980	0.893 / 0.854
Vogue	32	1.069 / 0.997	0.956 / 0.895	0.865 / 0.839
arXiv	8	1.479 / 1.454	0.859 / 0.851	0.697 / 0.698
arXiv	32	1.166 / 1.148	0.667 / 0.661	0.622 / 0.582
OpenAI-1536	8	1.316 / 1.131	0.851 / 0.825	0.641 / 0.620
OpenAI-1536	32	1.071 / 0.915	0.770 / 0.752	0.937 / 0.914
OpenAI-3072	8	1.032 / 0.953	0.798 / 0.798	0.721 / 0.702
OpenAI-3072	32	0.866 / 0.734	0.608 / 0.771	1.008 / 1.351

At batch 8/recall 0.97, CAGRA fp32 wins all four datasets under both estimators (JHQ ratios 0.641–0.893). At batch 32, ratios are 0.622–1.008; OpenAI-3072 favours JHQ under the measured threshold. Vogue's batch-8/0.95 lead requires interpolation. Grid resolution changes near ties.

Including qualifying endpoints, recall-0.90 ratios are 1.032–1.479 (batch 8), 0.866–1.166 (32), with three/one strict JHQ wins. Omitting endpoints overstates gains. Scheduling gains 1.7–3.3× over one block at 0.95, without ensuring cross-system wins.

Null-call medians are 0.012–0.072 ms and remain included. CAGRA uses Python, JHQ/RaBitQ C++; warm-ups differ (section 3.5.3).

7.3 Cartesian LUT, Packed Loads and Lane Refinement

The RQ2 ablations measure both kernel and application effects.

7.3.1 Exact table factorization and its throughput consequence

The LUT ablation tests the cost model in section 5.2 and FP32 ranking effects.

Figure 7.4 compares Cartesian table grouping on six datasets, probe counts 8/32/128/512, fixed $\alpha = 100$, and $g_{\mathrm{LUT}} \in \{1, 2, 4, 8\}$. Five rounds use batch 1,000 (Stella 999). Two groups lead in 22/24 cells. Exceptions are near ties: arXiv/128 favours one group by 0.23%; OpenAI-3072/8 favours four by 0.84%. Paired two-versus-one wins are 23/24; all dataset geometric means exceed one (1.021–1.318). Shared-index cells are not independent trials.

The two-group/one-group ratio ranges from 0.998 to 1.914. Four and eight groups each store 16 entries but require more lookups and additions. These arms show that smaller tables need not improve throughput; occupancy and instruction costs were not isolated. The maximum reported recall spread within any four-arm cell is 10^{-4}. At 128 probes and $\alpha = 100$, T4



Figure 7.4. Cartesian table grouping at fixed $\alpha = 100$ (v71). Group counts describe g_{LUT}, not parallel scan groups. Curves are medians of per-dataset QPS ratios to the one-group arm; shading shows their range.

compares identical candidates on Vogue, arXiv and OpenAI-3072. Full versus $16 + 16$ tables differ in 22.6–26.1% of candidate-score bit patterns, with maximum relative error 4.53×10^{-7}. Across all forms, 20.5–29.4% differ, with maximum relative error 4.64×10^{-7}. Vogue has 2–4/256 final-ID differences, while arXiv/OpenAI have none. Unchanged primary survivor sets can still produce different final IDs because the rounded primary score enters refinement (section 3.5.3).

7.3.2 Packed access and cooperative residual scoring

Packing wins 1.012–1.658× across 48 matched pairs (six datasets, four probe counts, $g_{LUT} \in \{1, 2\}$), with recall differences at most 10^{-4}. Both layouts are subspace-major. Packing/load/unpack change together. One aggregate per arm gives less timing evidence than five-round v71.

Figure 7.5 and table 7.5 then separate wider loads from lane cooperation in residual refinement. L0 is the original scalar path; L1 changes the load path without distributing a candidate across multiple lanes; L16 assigns 16 lanes to a candidate. The same process rotates arm order across five timing rounds, with three search calls per arm per round. Across twelve ordinary cells, L16 gains 1.015–1.789× overall. The separate high-budget arXiv/$\alpha = 200$ control reaches 2.383×.

Load-only L1 matches L0 bits yet gains 1.368× on arXiv. With lanes at $\alpha = 100$, refinement drops 4,277.5→716.8 μs (5.97×) versus 1.789× overall. OpenAI-3072 gains 8.81× in-kernel/1.171× overall. Other stages remain.

Table 7.5. Refinement at 128 probes. Ratios are paired end-to-end gains over L0, except the last column, which is the profiled refinement-kernel gain. The kernel and application gains must not be interchanged.

Dataset	α	$L1/L0$ QPS	$L16/L0$ QPS	Kernel gain
arXiv	100	1.368	1.789	5.97
BGE	32	1.026	1.062	1.97
OpenAI-1536	8	1.055	1.145	3.34
OpenAI-3072	4	1.063	1.171	8.81
Stella	16	1.030	1.053	2.66
Vogue	64	1.161	1.240	3.72

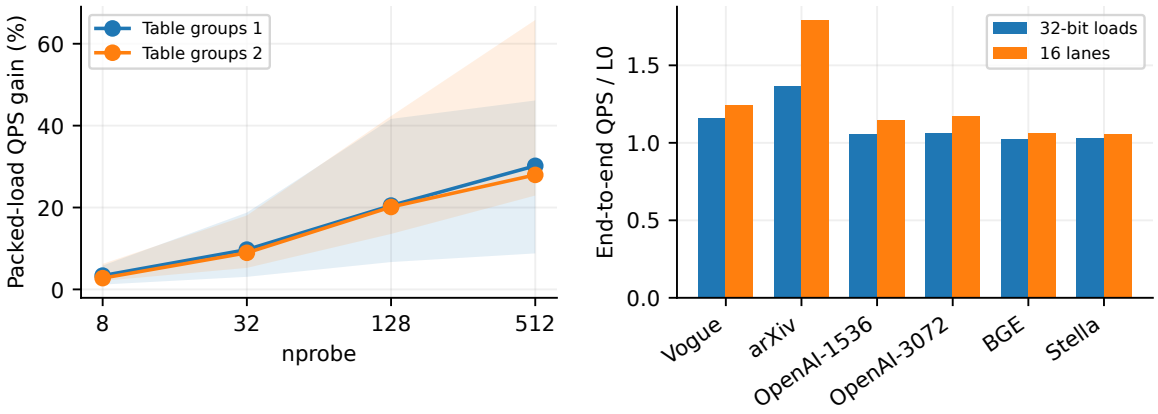


Figure 7.5. Packing and refinement controls. Left: median packed-load QPS gain across six datasets, with shaded minimum–maximum ranges, for each table grouping and probe count. Right: measured load-only and 16-lane end-to-end gains at 128 probes and the ordinary dataset-specific budgets. Panels use different experiment families.

For baseline wall-time fraction f and refinement speedup s, the replacement model gives

$$\frac{T'}{T} = (1 - f) + \frac{f}{s}, \qquad \frac{T}{T'} = \frac{1}{1 - f + f/s}.$$

ArXiv/OpenAI-3072 $f = 0.529/0.168$ predicts $1.787/1.175\times$, versus measured $1.789/1.171\times$. Separate profiling gives an explanatory estimate.

Lane FP32 reassociation reaches relative distance error $\approx 1.6 \times 10^{-6}$ and occasionally changes order. Sanitizer coverage is limited to recorded subsets; general bit identity is not claimed.

7.4 Global Top-c, Shared Bound and Probe Correctness/Ablation

7.4.1 Strict CPU reference for computed-score selection

T4 sorts GPU-produced scores by $(D_{\mathrm{P}}, \text{position})$ on CPU. The exporter follows scan layout/accumulation and matches valid returned scores bitwise; source comparison finds only export changes from the table-sweep version. Its 270 configurations cover Vogue/arXiv/OpenAI-3072, probes 1/8/128, $\alpha \in \{4, 100, 200\}$, $g_{\mathrm{scan}} \in \{0, 1, 2, 4, 8\}$ and both bound settings: 69,120 instances (table 7.6).

Table 7.6. T4 CPU-reference checks. Set and order equality use the complete primary key. Valid count checks $\min(c, n_c)$ survivors; final equality sorts retained computed refined scores by $(D_{\mathrm{H}}, \mathrm{ID})$, including their FP32 bits. Counts are query instances, with queries reused across configurations.

Path	Queries	Set / order exact	Tie-only	Valid count	Final exact
$g_{\mathrm{scan}} \geq 1$	55,296	55,296 / 55,296	0	55,296	55,296
$g_{\mathrm{scan}} = 0$	13,824	13,768 / 8,758	56	13,824	13,824

Table 7.6 confirms grouped full-key order and one-block distance order (56 tie-only set differences). All 19,120 underfull counts and 69,120 retained-score final selections pass, testing lemma 5.1 and propositions 5.1 and 5.2. Scan/bound and sanitizer coverage remains separate (section 3.5.3).

7.4.2 Shared bounds preserve the selection target

All 48 outputs agree; QPS ratios span 0.994–1.562×. At arXiv $\alpha = 200$, batch 1,000 and eight groups, sharing cuts merge 2,629→471 μs and scan 7,968→5,513 μs. Some cells regress slightly.

The pilot's first group holds 46–81% of winners. Equal quotas need fallback; unequal ones retain a bottleneck. Bounds avoid fixed quotas.

7.4.3 Exact probe selection and the limits of pipeline identity

Radix retains score arithmetic/OpenAI TF32 (section 7.2.3). All 1,792 synthetic cases pass. Probes and primary scores/positions match on all 1,000 arXiv/Vogue/OpenAI-3072 queries.

Table 7.7 shows end-to-end improvements of 1.034–1.167×. At batch 8, the profiled probe stage falls from 56.5 to 22.6 μs on arXiv, 52.7 to 14.7 μs on Vogue, and 77.8 to 16.7 μs on OpenAI-3072. Unchanged stages limit full-batch gains; each ratio uses its own baseline.

Vogue probe/primary dumps match bytewise on 1,000 queries across binaries (one-group/batch-8), yet final recall differs 0.9642/0.9644. The baseline alone also changes from 0.9642 to 0.9644 across identity/timing runs. Variation follows primary selection in this check and is unresolved. Within-process scheduled/one-block outputs match (section 3.5.1).

Optional FP16 probe sets match v69 on 936/979/998 of 1,000 arXiv/Vogue/OpenAI-3072 queries. High overlap does not establish exactness. Those results are excluded from the main E1 frontier and from the exact-probe claim.

7.5 Held-out Budget Quality, Throughput and Payback

RQ3 compares the policy of chapter 6 with fixed $\alpha = 100$ and the stronger universal control $\alpha = 64$.

7.5.1 Unlabelled selection and a labelled diagnostic oracle

The six datasets are tested at 128 and 512 probes with the fixed, sampled, FULL, enumeration and oracle controls in section 6.4. The disjoint pools and 64 draws per sample size follow section 3.2.

All rule variants use equation (6.4) with $k = 10$ and $\epsilon = 1$. FULL uses all calibration queries; the diagnostic oracle uses held-out labels (table 6.1).

7.5.2 Full-pool calibration as the quality-first working point

Full-pool calibration uses all 500 calibration queries once per cell. Its selected arms meet the held-out 10^{-3} loss target in 12/12 cells; the largest observed loss is 0.0006. Their rates are 0.908–1.000 of the GT oracle's, and calibration costs 9.5–60.3 ms. The geometric-mean rate ratio to fixed 100 is 1.111×, or 1.193× over the ten cells where fixed 100 also meets the target. Table 7.8 shows the complete comparison. Against fixed 64, full-pool selection is faster in six cells, equal in two and slower in four: its main advantage is the observed coverage of all twelve quality targets.

These cells favour FULL for quality and $S = 128$ for calibration cost (section 6.2).

Table 7.7. v69/v65 end-to-end throughput ratios for the exact computed-score probe change. Each workload–batch comparison has ten timing rounds per binary, collected in alternating runs. Ratios > 1 favour v69.

Dataset	Batch 8	32	256	1,000
arXiv	1.074	1.063	1.035	1.034
Vogue	1.109	1.103	1.052	1.041
OpenAI-3072	1.167	1.117	1.055	1.050

Table 7.8. Full-pool calibration ($S = 500$). Loss is relative to the best measured held-out recall; rates are kQPS from the selected fixed arm. "Cal." is one full-pool calibration call per cell, not a repeated stream timing. Held-out counts are 500, or 499 for Stella.

Dataset	Probe	α_{full}	Loss	Q_{full}	vs 100	vs oracle	Cal. ms
Vogue	128	100	0.0000	202.9	1.000	0.908	10.1
arXiv	128	200	0.0000	151.5	0.728	1.000	10.0
OpenAI-1536	128	8	0.0000	267.7	1.599	1.000	9.5
OpenAI-3072	128	4	0.0000	145.0	1.691	1.000	17.1
BGE	128	64	0.0006	124.9	1.075	0.932	17.1
Stella	128	16	0.0006	110.4	1.153	1.000	20.6
Vogue	512	100	0.0002	100.1	1.000	0.969	20.9
arXiv	512	200	0.0000	69.0	0.831	1.000	25.1
OpenAI-1536	512	8	0.0000	114.9	1.326	1.000	21.8
OpenAI-3072	512	4	0.0002	52.1	1.219	0.977	40.5
BGE	512	64	0.0006	50.8	1.033	1.000	42.3
Stella	512	16	0.0006	36.0	1.041	1.000	60.3

7.5.3 Lower-cost sampling and its quality risk

Table 7.9 reports the $S = 128$ result. The median selected budget matches the oracle budget in ten of twelve cells; it differs on arXiv at 128 probes and OpenAI-3072 at 512. Budgets span 4–200 across workloads. Median budget and mean recall need not describe one arm. Mean recall uses printed best recall minus mean loss, approximately to four decimals.

ArXiv has 40/64 and 32/64 loss crossings at 128/512 probes, despite mean loss 0.0009. Fixed 100 loses 0.0012/0.0016: better than default need not meet the target. BGE/512 has 24 crossings and $L_{95} = 0.0014$; Stella reaches 0.0030; OpenAI-3072 has none (table 7.9). Four losses can coexist with reported $L_{95} = 0$ (section 3.5.3).

7.5.4 Estimated throughput and calibration economics

With section 3.5.3's mean selected-arm estimate, $S = 128$ beats fixed 100 in ten cells (1.028–1.691×). ArXiv gives 0.908/0.917 with better mean recall: no throughput payback.

Against fixed 64, sampling wins 9/12 cells: geometric mean 1.054×, range 0.821–1.367× (OpenAI 1.052–1.367×). Both fixed controls fail arXiv quality, preventing a uniformly quality-matched comparison.

Table 7.9. Held-out quality at $S = 128$. Rule α is the median across 64 draws; oracle α uses held-out labels. $\overline{R}_{\text{rule}}$ is reconstructed mean recall. "Losses" counts draws with recall loss $> 10^{-3}$ relative to the best tested held-out recall; L_{95} is the logged lower-order 95th-percentile loss estimate. R_{100} is fixed-budget recall.

Dataset	Probe	α: rule/oracle	R_{100}	$\overline{R}_{\text{rule}}$	Losses	L_{95}
Vogue	128	64 / 64	0.9680	0.9677	5/64	0.0022
arXiv	128	100 / 200	0.9748	0.9751	40/64	0.0024
OpenAI-1536	128	8 / 8	0.9280	0.9279	4/64	0.0000
OpenAI-3072	128	4 / 4	0.9252	0.9252	0/64	0.0000
BGE	128	32 / 32	0.9198	0.9190	1/64	0.0008
Stella	128	16 / 16	0.9892	0.9884	6/64	0.0030
Vogue	512	64 / 64	0.9916	0.9915	2/64	0.0002
arXiv	512	200 / 200	0.9878	0.9885	32/64	0.0016
OpenAI-1536	512	8 / 8	0.9740	0.9738	4/64	0.0000
OpenAI-3072	512	4 / 8	0.9704	0.9702	0/64	0.0002
BGE	512	64 / 64	0.9670	0.9661	24/64	0.0014
Stella	512	16 / 16	0.9954	0.9945	7/64	0.0030

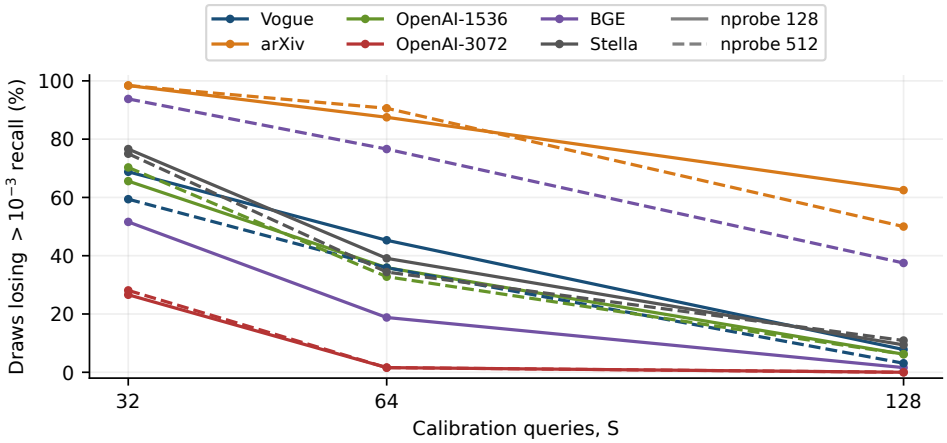


Figure 7.6. Held-out risk across sample sizes. The vertical axis is the percentage of calibration draws losing more than 10^{-3} held-out recall relative to the best tested budget. Increasing S also tightens the allowed slot-disagreement fraction because equation (6.4) keeps threshold one.

Table 7.10. Calibration economics, $S = 128$. Rates are kQPS and rule rate is mean selected-arm QPS. Ratios compare with fixed 100 and fixed 64. The "Batches" column gives payback relative to fixed 100 using 500 queries, or 499 for Stella; "Never" denotes no throughput payback. The quality column separately marks whether fixed 100 meets the held-out loss target.

Dataset	Probe	Q_{100}	$\overline{Q}_{\text{rule}}$	vs 100	vs 64	Cal. ms	Batches	Fixed 100 quality
Vogue	128	202.9	224.4	1.106	1.004	4.0	17.0	Pass
arXiv	128	207.9	188.8	0.908	0.821	4.2	Never	Fail
OpenAI-1536	128	167.5	268.5	1.603	1.367	3.9	3.5	Pass
OpenAI-3072	128	85.7	145.0	1.691	1.230	5.7	2.4	Pass
BGE	128	116.2	131.2	1.130	1.051	6.5	13.2	Pass
Stella	128	95.7	110.3	1.153	1.087	7.5	10.8	Pass
Vogue	512	100.1	102.9	1.028	0.996	7.5	55.4	Pass
arXiv	512	83.1	76.2	0.917	0.870	9.7	Never	Fail
OpenAI-1536	512	86.6	115.0	1.327	1.268	8.8	6.2	Pass
OpenAI-3072	512	42.8	52.1	1.219	1.052	13.1	6.2	Pass
BGE	512	49.2	51.1	1.039	1.006	15.8	41.2	Pass
Stella	512	34.6	36.0	1.041	1.028	21.3	37.1	Pass

Calibration costs 3.9–21.3 ms. Maximum positive estimated paybacks against fixed 100/64 are 0.54/2.72 s, or FULL 1.52/2.25 s (section 6.5, equation (6.15), and table 7.10). ArXiv controls fail quality; direct reuse below measures both cost and recall.

7.5.5 Complete decision traces and finite-pool risk

All 2,304 sampled and twelve FULL curves are monotone with enumeration-matched choices. Recorded subset sums reproduce 18,432 grid counts (equation (6.9)). This verifies recorded aggregates; arbitrary subset/position invariance and general distance-only nesting remain conditional (section 6.3).

Maximum loss is 0.0050 (arXiv/512), 0.0046 (BGE/512) at $S = 128$, and 0.0294 at $S = 32$. FULL stays within 0.0006.

ArXiv/128 predicted/observed crossings are 0.965/0.984, 0.875/0.875, 0.620/0.625 at $S =$ 32/64/128 (figure 7.7). The largest of 36 gaps is BGE/512: 0.144 (0.519/0.375). GT-loss prediction needs stable counts/held-out arm quality; calibration uses no labels.

Ten alternating repeats support a low-budget timing plateau, rather than general runtime monotonicity.

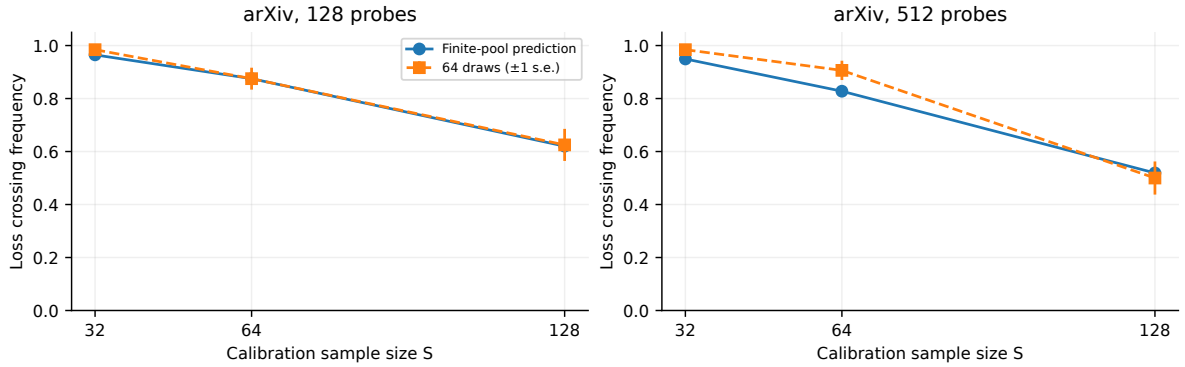


Figure 7.7. Finite-pool prediction and observed arXiv loss crossings. Bars show one estimated standard error over 64 independently drawn subsets conditional on the fixed pool, not uncertainty over future workloads.

7.5.6 Direct calibration-plus-reuse measurement

Four cells compare fixed 64/100, sampling and FULL across five rotated rounds of 500 repeated held-out batches. Timing includes gathering/calibration/budget installation/first graph, excluding build/warm-up/scan calibration (table 7.11).

Table 7.11. Direct total milliseconds for 500 reuse batches, averaged over five rounds, including budget calibration for sampled and FULL policies. Each batch has 500 queries. Quality is discussed with the timing results.

Dataset/probes	Fixed 100	Fixed 64	Rule $S = 128$	FULL
arXiv/128	1224	1102	1291	1680
OpenAI-1536/128	1491	1264	852	860
OpenAI-3072/128	2757	2160	1769	1783
BGE/512	5185	5018	4921	5062

Both OpenAI-3072 policies select 4 at recall 0.9252, matching fixed 100. OpenAI-1536 FULL/fixed 100 give 0.9280, sampling 0.9276. Table 7.11 gives times.

Direct savings start after 4/6 OpenAI-1536 batches and 4/9 OpenAI-3072 batches versus fixed 100/64 (figure 7.8); fractional fits differ. BGE sampling/fixed 100/FULL recall is 0.9658/0.9670/0.9664. ArXiv FULL selects slower 200 for quality; fixed controls fail.

7.6 Batch Regimes, G Scheduling and Workload Skew

7.6.1 Parallelism versus repeated selection work

Batch-8 eight-group gains are 2.06–2.76×; arXiv scan is 849→216 μs plus 16.7 μs merge. At batch 1,000 and $\alpha = 200$, independent grouping instead raises scan 3,530→7,652 μs plus 2,624 μs merging, reaching only 0.46× one-block QPS. Shared bounds reduce work, without removing this regime change.

7.6.2 A fixed rule evaluated on further configurations

The scheduler follows the fill/skew rule and pseudoquery calibration in section 5.5. Four configurations set the constants before the additional configurations were measured.

The means use thirteen small-batch configurations and eleven full-batch ones (Stella has 999 queries); held-out subsets use nine/seven (table 7.12). At batch 1,000, eight groups give 0.886×; worst regret is 1.217 (BGE/256).

Vogue's CV 0.338–0.622 selects four groups at batch 1,000. Ratios span 0.895–1.175; $(n_{\text{probe}}, \alpha) = (16, 16), (16, 64), (32, 64)$ give 0.955, 0.895, 0.955. CV measures spread, not total work: it enables the 0.97 lead yet overgroups low-probe loads (section 7.2).

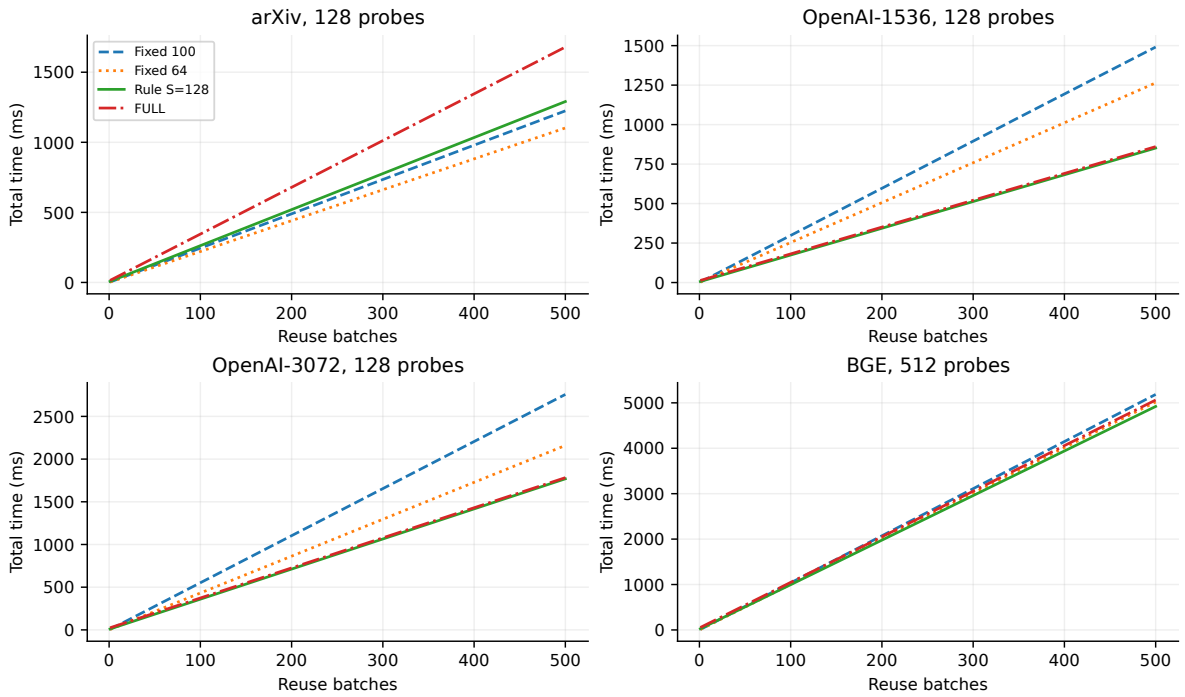


Figure 7.8. Direct mean cumulative time, including budget calibration and first-batch setup. Each curve replays one fixed held-out query pool. Policies can have different recall; timings alone are not a quality-matched ranking. Dataset/probe settings match table 7.11.

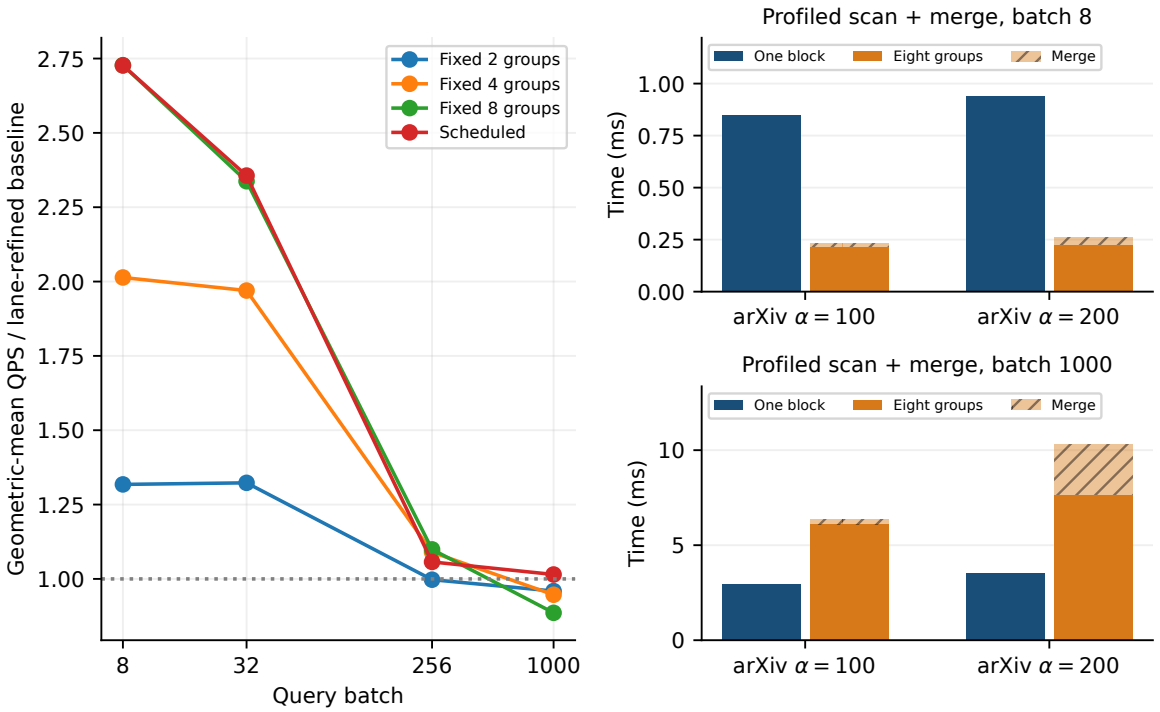


Figure 7.9. Execution regimes. Left: fixed shared-bound group counts and scheduled execution relative to the lane-refined one-block baseline (13 cells at small batches, 11 at batch 1,000). Right: paired scan and merge profiles at batches 8 and 1,000 for arXiv budgets 100 and 200. Hatching denotes merge time. The right panels use the independent-bound scan experiment; they explain a cost, not the scheduler's entire speedup.

Table 7.12. Integrated execution gains over v61 L16. Entries are geometric means of per-cell median paired speedups from five rounds. "Held out" excludes the four configurations used to set the rule; it means held-out configurations, not the budget experiment's held-out query split.

Batch	Cells	$g_s = 2$	$g_s = 4$	$g_s = 8$	Scheduled	Held out
8	13	1.318	2.014	2.728	2.727	2.930
32	13	1.323	1.970	2.337	2.356	2.560
256	13	0.997	1.090	1.099	1.057	1.041
1000	11	0.959	0.947	0.886	1.015	1.012

Across 250 variant/configuration pairs and 234,320 query comparisons, there are no non-tie primary differences or final-ID-set differences. Tied output order can differ. These are sequential synchronous throughput tests, not open-loop tail-latency tests.

7.7 Cold Construction, Resident Memory and Tested Scale

RQ1 is assessed through cold construction, resident memory and completed large-dataset searches; the E1 memory comparison also informs RQ4.

7.7.1 Cold builds and the lifecycle boundary

Three empty-cache repeats give Vogue 298–303 ms training and ≈ 0.56 s total, versus the slow archived 1.33 s. Stella takes 10.54–10.65 s versus archived 10.40 s (table 7.13).

Sections 3.3 and 3.5.2 define startup/build boundaries; figure 7.10 compares repeated builds.

Stella takes 74.55/82.45 s with isolated CAGRA int8 (quantisation included), and median 86.59 s across three IVF-PQ configurations, versus JHQ's 10.54–10.65 s. Quality/resident memory are separate comparisons.

Five arXiv builds span 0.0012 recall. Stella startup takes 18.2–18.6 s with warm pages, 129.9 s on first read. Sampled 20 ms peaks are BGE 26.08 GiB and Stella 22.58 GiB. BGE's 12 GiB assignment buffers precede its 23.05 GiB gather payload, so they must not be summed (equation (4.13)).

7.7.2 Resident allocation and where compression helps

Figure 7.11 combines the component model in section 4.4 with measured resident totals. Its unattributed remainder is retained, and the preceding construction measurements give the sampled peaks.

Residuals occupy 80.7% of BGE's 11.93 GiB and 82.9% of Stella's 20.44 GiB totals, including context/workspace. Raw vectors need 38.50/67.81 GiB. Smaller α saves reads, not stored residual bytes (section 4.4).

Table 7.13. Cold JHQ construction and separately measured resident memory. Train/add are component medians of three supplementary normal builds; Total is the median paired sum. Memory comes from v71's two-group, 128-probe, $\alpha = 100$ runs and includes context and workspace. "Residual" is the modelled residual-code share of the measured used total, not the fraction read per query.

Dataset	Train s	Add s	Total s	Used GiB	Raw GiB	Residual
Vogue	0.30	0.26	0.56	1.47	2.67	45.4%
arXiv	0.48	0.51	0.99	2.58	6.45	62.5%
OpenAI-1536	0.61	0.38	1.00	2.39	5.72	59.7%
OpenAI-3072	1.16	0.67	1.82	4.14	11.43	69.0%
BGE	3.56	3.65	7.20	11.93	38.50	80.7%
Stella	3.55	6.99	10.55	20.44	67.81	82.9%

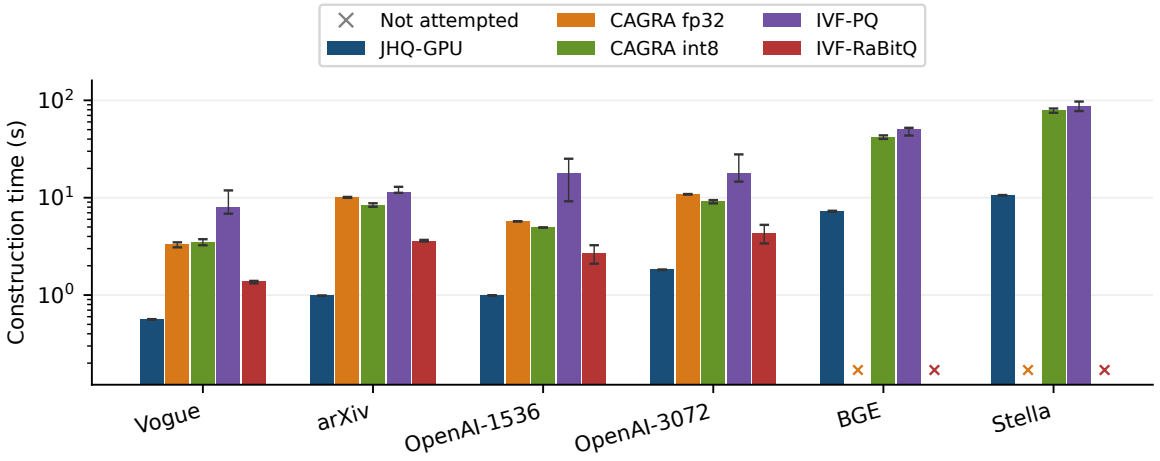


Figure 7.10. Recorded construction costs, with CAGRA int8 included. JHQ is the median and range of three supplementary normal cold builds with warm input pages; other bars are medians across successful measured index configurations, with minimum–maximum whiskers. Int8 includes dataset quantization; RaBitQ excludes the required serialization round trip. Missing large-dataset bars denote unattempted configurations, not measured build failures. The vertical axis is logarithmic.

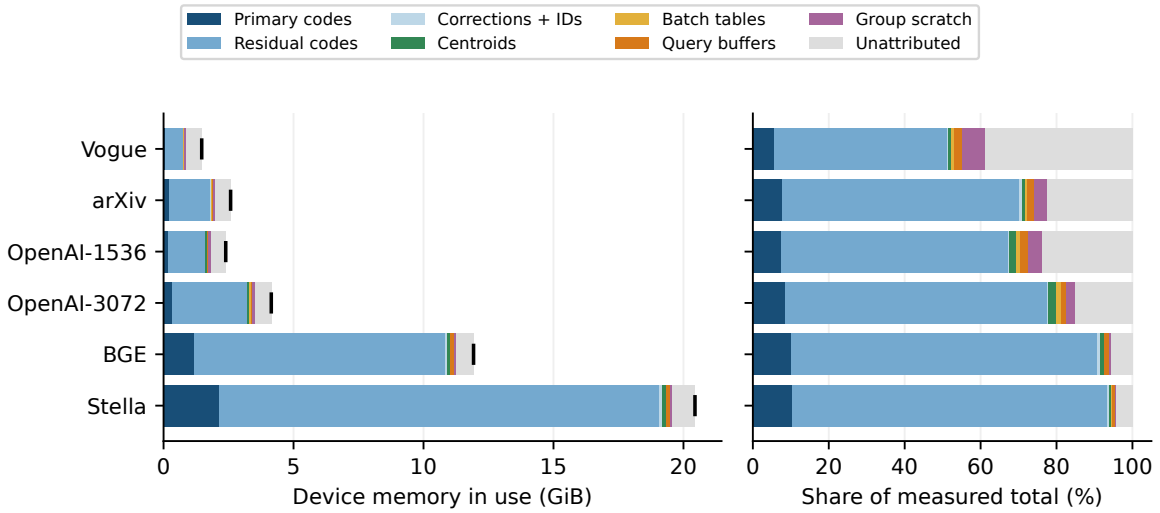


Figure 7.11. Resident-memory decomposition at two-group tables, 128 probes and $\alpha = 100$. Component sizes are modelled, markers are measured used totals, and hatching is the unattributed remainder. The totals include context and workspace. The preceding construction measurements give sampled peaks; these resident markers are not peaks.

The E1 measurements provide a separate cross-system view in table 7.14. Subtracting the baseline-measured 507.2 MiB context from JHQ's process total gives approximately 0.29–0.39 of CAGRA fp32's index allocation. JHQ's figure includes workspace, whereas CAGRA's is the build-time index delta. These are parameter-sweep ranges, not memory-matched recall points or a comparison of equal peak budgets. The CAGRA int8 allocations in the same table are of a similar order to JHQ's: compression relative to fp32 is shared by competing designs. The component model in table 4.1 explains this comparison.

7.7.3 Large-dataset feasibility and competitive regions

BGE contains 10,091,524 vectors and Stella 17,776,615, both with 1,024 dimensions and 32,768 IVF lists. Stella's queries are sampled from the corpus; its result therefore characterizes that benchmark rather than an independently collected external-query distribution. Figure 7.12 includes successful CAGRA int8 and IVF-PQ measurements. Full-precision CAGRA/RaBitQ/FP32 brute force were excluded where predicted storage exceeded capacity. These are planned exclusions; no out-of-memory or recall failure was measured.

CAGRA int8 fits Stella. On BGE it leads 1.35–2.96× at recall 0.88–0.95; JHQ leads 1.086× at int8's maximum 0.9579. Maximum BGE/Stella recall: int8 0.9579/0.9783, JHQ 0.9663/0.9952, PQ 0.9525/0.9896.

Stella gives JHQ 185.4 kQPS at 0.978679 recall versus int8 66.3 at 0.978278. The 2.80× point ratio is not recall-matched; int8 lacks upper-region coverage (figure 7.12).

7.8 Numerical Robustness, Anomalies and Recall Attribution

7.8.1 Quality resolution and repeated execution

A single GT hit changes Recall@10 by

$$\Delta r = \frac{1}{nk} = \begin{cases} 10^{-4}, & n = 1000, \\ 2 \times 10^{-4}, & n = 500. \end{cases}$$

Stella's $n = 499$ step is slightly larger. This step size is not an uncertainty interval.

All 369 E1 baseline repeat-ID flags stay, including tied reordering. The 401 convention adds cross-protocol differences (section 3.5.2); these are not independent builds.

Near 0.97, local slopes imply 1.01–3.17% QPS change/10^{-3} recall before snapping. Five-build recall ranges are Vogue/arXiv/OpenAI-1536/3072: 0.0008/0.0012/0.0004/0.0002. This suggests local sensitivity, not a retrained-frontier bound. Cached-model FP32 checks retain OpenAI's advantage (section 7.2.3); arXiv/LUT near ties remain inconclusive.

RaBitQ's median five-round ranges are 3.79%, 3.34%, 2.87%, 5.25% of median QPS on Vogue/arXiv/OpenAI-1536/3072. These spreads are not confidence intervals. To reduce occasional 30–60 ms calls' influence, we recompute the 1,000-query rate as 10^6/ median(call ms). At recall 0.97, the resulting JHQ/RaBitQ ratios are 1.063, 0.995, 1.118 and 1.100. The three favourable directions and arXiv near parity remain, so the result is not solely a consequence of long calls.

Table 7.14. E1 allocation ranges. JHQ is process total minus a 507.2 MiB context estimate, including query workspace; both CAGRA columns are index allocations measured during build. Ranges cover the parameter sweep.

Dataset	JHQ net process GiB	CAGRA fp32 index GiB	CAGRA int8 index GiB
Vogue	0.90–1.01	2.89–2.91	0.89–0.91
arXiv	2.01–2.19	6.84–6.98	2.01–2.15
OpenAI-1536	1.79–2.33	5.95–5.97	1.67–1.68
OpenAI-3072	3.54–3.82	11.67–11.68	3.10–3.11

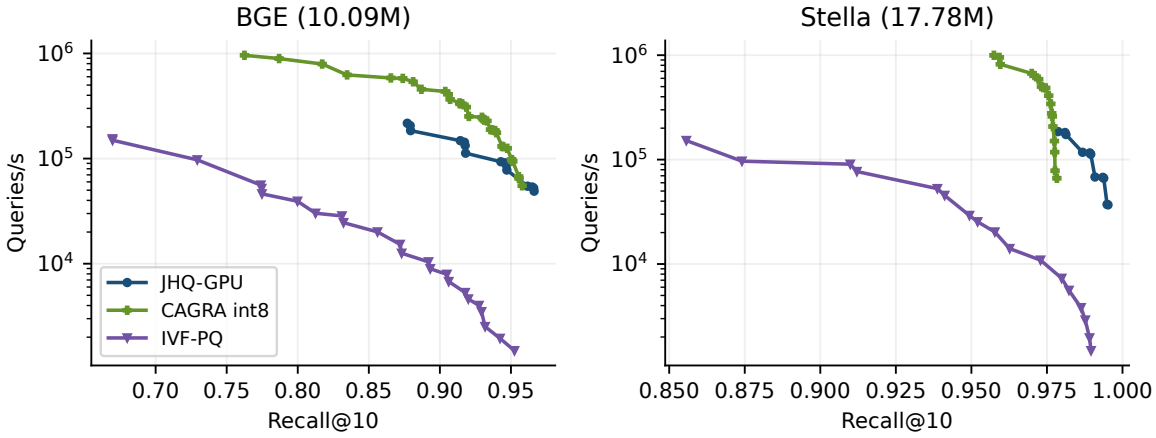


Figure 7.12. Large-dataset host-to-host frontiers, including the successful isolated CAGRA int8 reruns. BGE/Stella use 1,000/999 queries. Endpoints mark tested-grid coverage.

7.8.2 Exactness is attached to a particular stage

The stage contracts in table 5.1 distinguish computed-score selection from raw-space kNN; lane rounding, cutoff ties and process variation qualify different stages.

The archived reconstruction pilot measures the remaining gap to full-probe reconstructed scoring. Across seven workload–budget arms of 128 queries each, scoring all probed candidates in FP64 improves recall over GPU outputs at selected budgets by 0–0.002344. For arXiv at 128 probes, $\alpha = 200$ matches the full-IVF reconstructed answer set on all 128 queries. At 100, the recall is 0.9734375 versus 0.9750 for full-IVF scoring: an additional 0.0015625 budget deficit beside the joint routing/reconstruction deficit of 0.0250 (section 6.2). These pilot aggregates include two budgets on one dataset–probe workload. FP64/GPU arithmetic differences prevent pure budget attribution in every arm. Separating routing/representation needs stage-wise GT coverage.

8 Discussion and Future Work

8.1 Three Cross-Layer Insights

8.1.1 A useful representation exposes a useful unit of reuse

More reuse does not help Mailman; two-group LUTs usually win (figure 7.4). Codebook quality must therefore be evaluated with achievable table placement/access cost (sections 5.2 and 5.3).

8.1.2 Useful parallelism is bounded by duplicated state and selection

Once queries supply enough parallelism, staging/selection/merge overhead can dominate (figure 7.9). Vogue/BGE expose missing absolute-work/decomposition features (sections 7.2 and 7.6). Richer scheduling must repay calibration; bounds alone are not isolated.

8.1.3 Execution speed changes the value of avoiding refinement

Assume fixed $T_0 > 0$, refinement times $F_h \geq F_\ell > 0$ and common multiplier $\eta \in (0, 1]$. The low-budget speedup and its derivatives are

$$S(\eta, T_0) = \frac{T_0 + \eta F_h}{T_0 + \eta F_\ell}, \tag{8.1}$$
$$\frac{\partial S}{\partial \eta} = \frac{T_0(F_h - F_\ell)}{(T_0 + \eta F_\ell)^2} \geq 0, \qquad \frac{\partial S}{\partial T_0} = \frac{\eta(F_\ell - F_h)}{(T_0 + \eta F_\ell)^2} \leq 0.$$

For $F_h > F_\ell$, faster refinement reduces the value of cutting its budget; lower fixed cost increases it. The derivatives hold with non-refinement work fixed. Budget-dependent scan maintenance and resource transitions require the controlled comparisons of sections 7.5 and 7.5.6. Quality and reuse remain constraints (table 7.8 and figure 7.6).

8.2 Two Deep Negative-Result Case Studies

8.2.1 Mailman-style shared sums: arithmetic reuse meets scatter cost

Mailman reuses finite-alphabet sums [20]. Frozen Vogue/arXiv/OpenAI-3072 streams compare atomic buckets/private histograms at widths 4/6/8 with packed two-table scoring: 128 probes, batches 8/1,000, fixed codes/mathematical scores.

The best alternatives reach only 0.187–0.581 of baseline QPS (figure 8.1). Biased bucket destinations dominate profiles; private histograms add state/shared updates. Narrower groups sometimes help, but none wins. Other finite-alphabet designs remain open.

Only scores were tested, so cutoff checks do not establish neighbour-pipeline equivalence. The replacement model gives

$$\frac{T'}{T} = 1 + f\left(\frac{1}{s} - 1\right) > 1 \quad (0 < f \leq 1,\ 0 < s < 1).$$

This model is not measured end-to-end performance; integration stopped. A dictionary study also stopped at structural statistics. Neither contributes a system-frontier point.

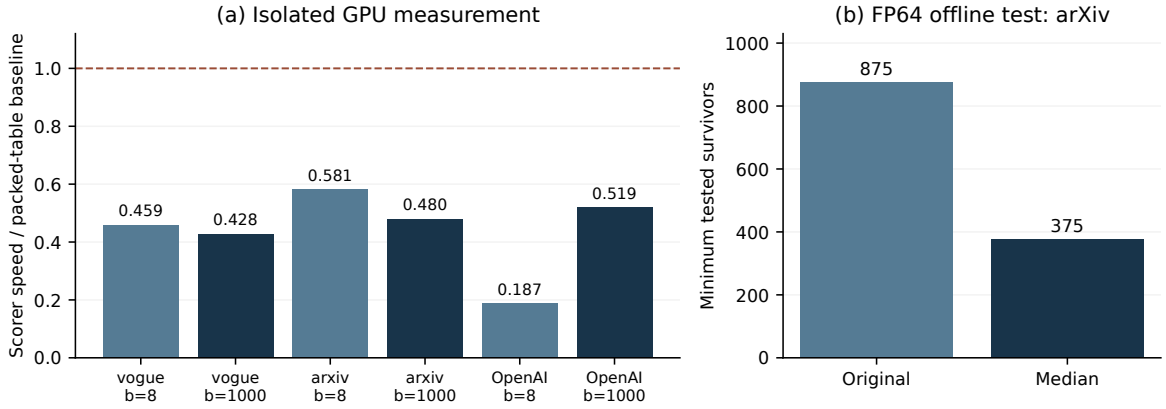


Figure 8.1. Two distinct evidence boundaries. Left: best measured Mailman scorer / packed-table scorer speed on an RTX 5090, with Vogue, arXiv and OpenAI-3072 at two batches and 128 probes; all ratios are below one. Right: minimum survivor counts meeting the original JHQ recall target on arXiv's 400-query FP64 offline test split. The candidate reduction is not a GPU speedup.

8.2.2 Rank-oriented coding: a local gain is not a general replacement

FP64 candidate replay uses 400/200/400 calibration/validation/test queries. It tests weighted scores, median/list thresholds, learned losses and linear-MMSE decoding. Lower MSE can worsen neighbour order; primary-level changes also move residual reference points.

Median thresholds reduce arXiv's minimum tested qualifying count 875→375; two other workloads fail. For orthogonal Π,

$$\bar{y} = \frac{1}{N}\sum_i \Pi x_i = \Pi \bar{x}, \qquad \|\bar{y}\|^2 = \bar{x}^\top \Pi^\top \Pi \bar{x} = \|\bar{x}\|^2.$$

Rotation therefore preserves mean-vector norm rather than removing bias. Centring must also preserve the refined-score ceiling.

Median-level GPU follow-up retrains residuals. ArXiv's original $\alpha = 100$ target gains 1.40/1.21× at batch 8/1,000; Vogue reaches 0.9630 below 0.9644, OpenAI-3072 ties. Three rounds use separate states/queries, outside E1.

8.3 Threats to Internal and External Validity

Table 8.1 consolidates threats to the main interpretations. Controls are in Chapter 3; repeat/build/precision checks are in section 7.8.

Table 8.1. Threats that can reverse or narrow a conclusion.

Claim at risk	Alternative or reversal	Control and remaining test
Execution gain	Changed scores, ties or index state explain an apparent speedup	Strict T4 CPU sorting and table-form diagnostics; finite coverage, with candidate dumps no longer retained
Budget quality	Finite-pool sampling or drift hides damaging queries	Disjoint GT evaluation, full traces, subset-sum checks and FULL enumeration; future-query and arbitrary-subset invariance remain untested
Competitive margin	Grid density, call tails or API overhead changes the ordering	Both recall estimators, retained slow calls and int8 baseline; TF32/FP32 sensitivity; T5 exposes small-batch reversals; cross-device comparisons remain unmeasured
Capacity/build claim	Transient allocations exceed a resident sample	Phase bytes, repeated empty-cache builds and 20 ms device peaks; sub-interval allocations and full retrained frontiers remain unmeasured

Evidence covers one GPU, six prepared datasets and squared-L2 top-10 (section 7.2.6). Near duplicates may remain; relevance labels are absent. Updates, concurrency, drift, service latency, billion-scale and multi-device use are untested.

8.4 Testable Research Directions

8.4.1 Risk-controlled agreement with a reuse-aware objective

Future cost-aware calibration needs valid query-level bounds, matched-regime counts, fixed reference, sampling/order assumptions and drift renewal. Absolute GT risk additionally needs labels or a justified reference-to-GT link.

Compare sampling/rank reuse/cost-aware choice by held-out total time and tail agreement. A four-cell sequential controller lost to extra launches. A replacement must beat a quality-matched control while meeting its risk bound.

8.4.2 Capacity-aware residual placement

A future residual cache could fetch host blocks under a device cap. Fix representation, routing, survivors and arithmetic; vary cap, batch and locality; measure bytes, hits, overlap, tails and QPS. This capacity study is unimplemented; lowering α does not relocate stored residuals.

8.5 Conditional Deployment Recommendation

For recall 0.97, E1 supports batch-1,000 throughput; at batch 8, T5 favours CAGRA fp32. Batch-32 OpenAI-3072 depends on the estimator (section 7.2.6). Begin with FULL, then validate quality/reuse cost/memory (section 8.3).

9 Conclusion

9.1 Answers to the Four Research Questions

RQ1: Construction and capacity. GPU construction produces a searchable resident hierarchy. The largest test holds 17.78 million vectors at 20.44 GiB, built in 10.54–10.65 s; section 7.7 reports construction peaks.

RQ2: Correct and efficient execution. Cartesian splitting preserves real-arithmetic scores; complete-key retention/merge and witnessed bounds preserve grouped selection under their premises. Two-group LUTs win 22/24 cells. Scheduling gains 2.727/1.015× at batch 8/1,000 against lane refinement (sections 7.3 and 7.6). Strict CPU checks confirm grouped order; one-block differences are cutoff ties (section 7.4.1).

RQ3: Refinement budget and quality. FULL calibration meets the tested relative recall-loss target in 12/12 cells, with maximum observed loss 0.0006. $S = 128$ reaches 40/64 crossings in one cell (section 7.5). FULL matches enumeration; ten distance-only cells lack general nesting. Reference agreement is not population GT recall. Four direct reuse tests show workload-dependent quality/time gains.

RQ4: Competitive operating regions. At batch 1,000/recall 0.97, JHQ leads on three datasets and nearly ties RaBitQ on arXiv; Vogue's lead needs grouping (section 7.2). OpenAI E1 uses TF32. FP32 checks retain gains on the TF32-trained cache (section 7.2.4). Int8/RaBitQ win other recall regions; CAGRA fp32 wins all batch-8 datasets. Batch-32 near ties depend on the estimator (section 7.2.6).

9.2 Closing Principle and Outlook

Representation, execution and effort must share a quality target. Future work targets calibrated reuse and residual placement (section 8.4).

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